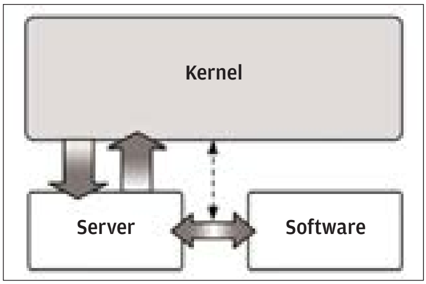
loads all the kernel modules even if many of them are not need—for example, loading LAN and network management modules on a home computer that isn't connected to a network.

The UNIX and Linux kernels are classic examples of monolithic kernels. The Linux and BSD (Berkley Software Distribution)—a variant of UNIX—kernels can even load new modules while it is running in the system's memory to extend their capabilities when needed.

1.2.3 The Team Captain —The Microkernel

The microkernel is a bare-bones kernel that only takes care of very critical functions such as basic hardware abstraction, managing processes, and handling communication between processes. All other functions are carried out by separate applications called *servers*, which are run in the user mode.



The Microkernel - everything is done through a server

The advantage of this design was supposed to be efficiency—servers would be loaded into the system's memory only when needed, as opposed to the monolithic approach which loaded even unnecessary modules into memory. Moreover, a badly-written server would only cause itself to crash without crashing the entire system. However, the kernel has to manage servers like regular processes—switching between them to give each of them access to resource—and the time consumed in this activity caused it to slow down. By the mid-90s, researchers had all but given up on try-

Microkernel vs. Monolithic

"Asking the system for the time"
Time taken (Monolithic UNIX Kernel):
20 Microseconds
Time taken (Mach 3 Microkernel):
114 Microseconds